Title: HR Analytics – Employee Attrition Prediction

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1. Introduction

Employee attrition is a major issue for organizations. This project analyzes employee data to understand the factors leading to attrition and builds a predictive model to identify employees likely to leave.

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2. Objectives

- Analyze employee attrition patterns
- Identify key factors influencing attrition
- Build a machine learning model
- Provide actionable business insights

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3. Dataset

The dataset used is IBM HR Analytics dataset from Kaggle.

It includes:

- Age
- Department
- Salary
- Job Role
- OverTime
- Years at Company
- Attrition

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4. Tools & Technologies

- Python (Pandas, NumPy, Matplotlib, Seaborn)
- Scikit-learn
- Power BI
- Jupyter Notebook

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5. Methodology

Step 1: Data Preprocessing

- Handled categorical variables
- Encoded text data

Step 2: Exploratory Data Analysis

- Attrition count
- Salary vs attrition
- Department analysis

Step 3: Model Building

- Algorithm: Random Forest
- Data split: 80% training, 20% testing

Step 4: Evaluation

- Accuracy score
- Confusion matrix

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6. Results

- The model achieved good accuracy
- Employees working overtime have higher attrition
- Lower salary employees tend to leave more
- Early-stage employees are more likely to resign

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7. Dashboard

A dashboard was created using Microsoft Power BI Desktop showing:

- Attrition rate
- Department-wise analysis
- Salary trends
- Key influencing factors

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8. Conclusion

This project helps organizations understand employee behavior and take proactive measures to reduce attrition.

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9. Future Scope

- Use advanced models like XGBoost
- Deploy model using web applications
- Integrate real-time HR systems

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10. References

- Kaggle Dataset
- Scikit-learn Documentation
- Power BI Documentation